

# Photo-Realistic Streamable Free-Viewpoint Video

Shaohui Jiao  
Yuzhong Chen  
Zhaoliang Liu  
jiaoshaohui@bytedance.com  
chenyuzhong.yui@bytedance.com  
liuzhaoliang.42@bytedance.com  
Bytedance Inc.  
Beijing, China

Danying Wang  
Wen Zhou  
wangdanying.a@bytedance.com  
zhouwen.me@bytedance.com  
Bytedance Inc.  
Beijing, China

Li Zhang  
Yue Wang  
lizhang.idm@bytedance.com  
wangyue.v@bytedance.com  
Bytedance Inc.  
Beijing, China

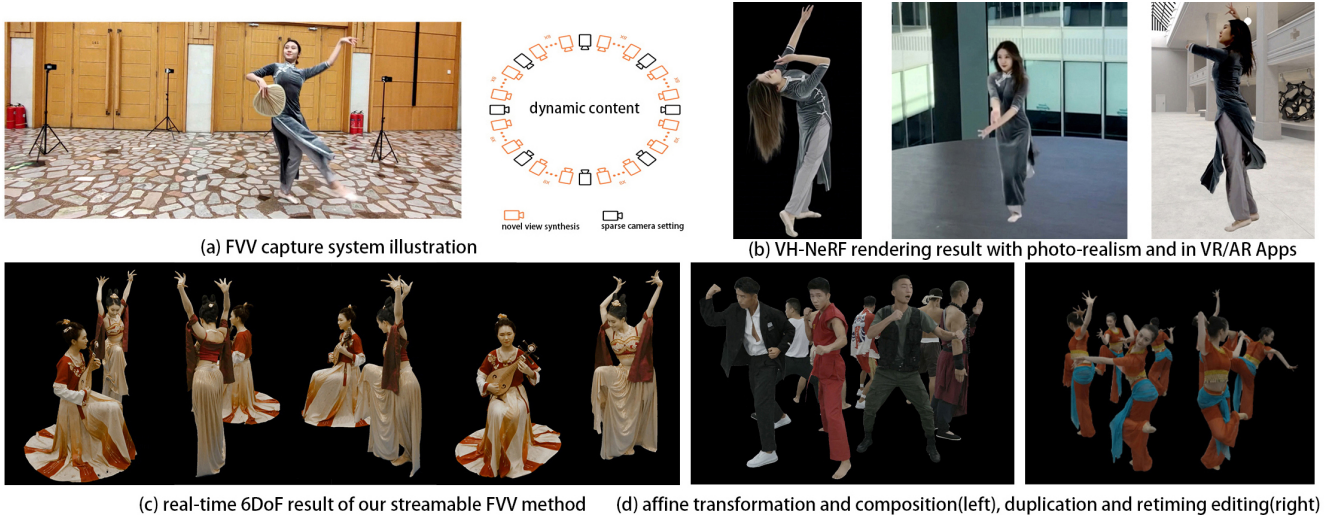


Figure 1: Free-Viewpoint Video Capture System and Photo-Realistic Rendering Results with Various Editing.

## ABSTRACT

We present a novel free-viewpoint video(FVV) framework for capturing, processing and compressing the volumetric content for immersive VR/AR experience. Compared to previous FVV capture systems, we propose an easy-to-use multi-camera array consisting of mobile phones with time synchronization. In order to generate photo-realistic FVV results with sparse multi-camera input, we improve the novel view synthesis method by introducing visual hull guided neural representation, called VH-NeRF. Our VH-NeRF combines the advantages of both explicit models by traditional 3D reconstruction and the notable implicit representation of Neural Radiance Field. Each dynamic entity's VH-NeRF is learned and supervised by the visual hull reconstructed data, and can be further edited for complex and large-scale dynamic scenes. Moreover, our FVV solution can do both effective compression and transmission on multi-perspective videos, as well as real-time rendering

on consumer-grade hardware. To the best of our knowledge, our work is the first solution for photo-realistic FVV captured by sparse multi-camera array, and allow real-time live streaming of large-scale dynamic scenes for immersive VR and AR applications on mobile devices.

## ACM Reference Format:

Shaohui Jiao, Yuzhong Chen, Zhaoliang Liu, Danying Wang, Wen Zhou, Li Zhang, and Yue Wang. 2023. Photo-Realistic Streamable Free-Viewpoint Video. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Posters (SIGGRAPH '23 Posters)*, August 06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3588028.3603666>

## 1 INTRODUCTION

With the rise of virtual reality(VR) and augmented reality(AR), immersive video has become a hot research topic in both academic and industrial communities, since it provides higher degree of freedom(DoF) than traditional video offers. One type of the immersive video is free-viewpoint video, which allows the viewers interactively choose their viewpoints in 3D space to observe the dynamic scenes from arbitrary perspectives. However, compared to traditional video pipeline, free-viewpoint video has challenges in both acquisition and processing, as well as the content delivery pipeline.

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SIGGRAPH '23 Posters, August 06–10, 2023, Los Angeles, CA, USA

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ACM ISBN 979-8-4007-0152-8/23/08.

<https://doi.org/10.1145/3588028.3603666>

### 1.1 Capture System

Immersive video is usually captured by multi-view cameras, which are expensive. To tackle the problem, we employ a limited number of handheld consumer cameras to capture volumetric content.

For our capture system, only around 12 mobile phones are used, which are well-synchronized by the implementation of the previous work of [Akhmetyanov et al. 2021], along with software engineering optimizations. The synchronization tests indicate that our multi-camera array can stream videos at 30fps within 2ms of time difference and the intra-array temporal drift is less than a video frame time lasts for 1 hour. Our proposed capture system, although contains a very few number of cameras, is demonstrated to be sufficient to render novel views and generate high quality volumetric content, as shown in Fig 1.

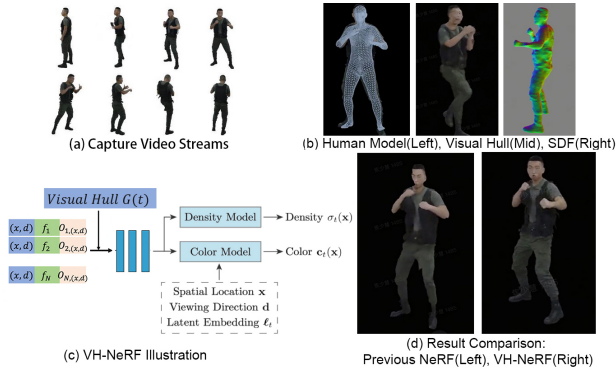


Figure 2: VH-NeRF illustration and comparison

### 1.2 VH-NeRF Method

For free-viewpoint video generation of human-centric dynamic scenes, traditional model-based solutions rely on high-fidelity 3D reconstruction and texture rendering in novel views. However, they may still have difficulties to render real-life results. On the other hand, image based rendering approaches generate novel views directly from multi-view, time-synchronized videos. Recent progress of neural rendering techniques bring huge potential for photo-realistic novel view generation. The notable approach Neural Radiance Field (NeRF) [Mildenhall et al. 2020] and further progress on editing feature [Zhang et al. 2021] combine implicit representation with volumetric rendering to achieve compelling novel view synthesis with rich view-dependent effects. However, sparse camera settings are still crucial for neural representation technologies since they lead to information missing due to both occlusions and view foreshortening, that causes challenges of novel-view synthesis in large-scale and complex dynamic scenes.

In order to render high-quality results and support large-scale dynamic scenes, our method combines model reconstruction methods with the state-of-the-art novel-view synthesis technique of NeRF. The core of our approach is a new visual hull guided neural representation, called VH-NeRF, where each dynamic entity’s neural radiance representation is associated with a learnable geometric descriptor, supervised by the visual hull reconstructed data. The

proposed method can handle visibility issues and achieve a relatively sharper depiction of volumetric content with complete novel view synthesis result with photo-realism. Moreover, our proposed method reduces memory requirements since VH-NeRF can trim around more than 80% of the data during computation.

After each VH-NeRF is learned for individual dynamic entity, a geometry-aware volume rendering scheme is further introduced to assemble all the dynamic entity’s neural representations, for completing the large-scale dynamic scenes. It is noted that our VH-NeRF method enables various editing features for free-viewpoint video, like overlay composition from different dynamic entities, duplication and affine transformation, as well as temporal editing like re-timing performers’ movements, demonstrated in Figure 1.

### 1.3 Video Compression and Streaming

Live streaming of free-viewpoint video is a great challenge over bandwidth-limited network connections. In order to fit in the current MPEG video codec standard, we represent the learned VH-NeRF as multi-perspective video streams consisting of both captured streams and view synthesis streams, which exhibits a lot of self-similarities. The redundant pixels are removed for view pruning to obtain patched atlases as compressed video. And viewing parameters and coarse geometry information are coded as well within extension information along with compressed videos.

Video switching technology via RTC (Real-time Communications), with latency less than 100ms, is used to shift the view-dependent perspectives correctly corresponding to users’ changing positions and observing view parameters. For real-time applications, novel views at interval less than 7 degrees are interpolated based on decoded geometry information on users’ devices. With both video switching technology and novel view interpolation implementation, users can get correct perspective views at 30fps with bandwidth within 30Mbps, which can be demonstrated by screen recording of VR headsets in the accompanied video.

## 2 CONCLUSION

Our contributions are as following:

- Multi-camera time-synchronization established on mobile phones, contributing to an easy-to-use capture system.
- VH-NeRF is proposed, achieving photo-realistic view synthesis result by combing the advantages of explicit geometry and implicit representation. Moreover, editing is supported to generate more complex and large-scale dynamic scenes.
- Immersive video compression and state-of-art transmission technology, accompanied with real-time view interpolation on users’ mobile devices, make a streamable FVV system with low latency within consumer-grade bandwidth.

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